LOCAL COHOMOLOGY DESCRIBED AS A COLIMIT OF EXT GROUPS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The description of local cohomology through the torsion of a K-injective representative. Used inside the counted proof (source lines 1322-1326). Writes local cohomology as the torsion of a K-injective complex.
- The statement that a filtered colimit of Hom complexes represents the corresponding hom-colimit. Invoked inside the counted proof (source lines 1327-1329) as a cross-chapter pointer. Turns the colimit into the hom-colimit identity.
- The identification of the cohomology of the Hom complex with Ext groups, and exactness of filtered colimits of abelian groups. Invoked at the end of the counted proof (source lines 1330-1333). Rewrites the cohomology of the colimit as the colimit of Ext groups.

Source reference. Original reference derived-lemma-colim-hocolim: sources/stacks-dualizing/pinned-original-source.tex, source lines 1328–1328. Navigation pointer only: the counted text refers to the Stacks item derived-lemma-colim-hocolim, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a ring and let $I \subset A$ be a finitely generated ideal. For any object K of $D(A)$ we have*

$$R\Gamma_I(K) = \mathit{hocolim}\, R\mathrm{Hom}_A(A/I^n, K)$$

in $D(A)$ and

$$R^q\Gamma_I(K) = \mathrm{colim}_n \mathrm{Ext}^q_A(A/I^n, K)$$

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as modules for all $q \in \mathbf{Z}$.

Proof. Let $J^\bullet$ be a K-injective complex representing K. Then

$$R\Gamma_I(K) = J^\bullet[I^\infty] = \operatorname{colim} J^\bullet[I^n] = \operatorname{colim} \operatorname{Hom}_A(A/I^n, J^\bullet)$$

where the first equality is the definition of $R\Gamma_I(K)$. By Derived Categories, Lemma derived-lemma-colim-hocolim we obtain the first displayed equality in the statement of the lemma. The second displayed equality in the statement of the lemma then follows because $H^q(\operatorname{Hom}_A(A/I^n, J^\bullet)) = \operatorname{Ext}^q_A(A/I^n, K)$ and because filtered colimits are exact in the category of abelian groups. $\square$